Algorithmic Empathy: A Qualitative Study of Player Experience of Emotionally Adaptive Game Narratives

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Master's Degree Project in
Design for Creative and Immersive Technology
HT2025/VT2026
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Abstract

Introduction Emotionally adaptive games adjust their output in response to player behaviour, yet the existing literature relies almost exclusively on physiological signals, questionnaires, and performance metrics to evaluate their effects. No published study has used in-depth qualitative interviews to investigate how players subjectively experience emotional adaptation, nor has any study compared how different modes of adaptation shape that experience within the same game.

Research Question How do players experience emotionally adaptive game narratives when the mode of adaptation varies?

Method A purpose-built text-based interactive fiction game, Fabricade, served as a technology probe implementing two adaptation conditions: Condition A (Dialogue) adapted narrative dialogue and character response, while Condition B (Atmosphere) adapted atmospheric presentation through visual overlays, text glitch effects, and reduced system narration. Ten participants each played both conditions in a counterbalanced within-subjects design, followed by a semi-structured interview. Interviews were analysed using reflexive thematic analysis. A structured questionnaire and session log data provided supplementary triangulation.

Results Five themes were identified. Players experienced fabricated emotions as genuinely affecting regardless of their awareness that the content was not real. Atmospheric adaptation was consistently perceived as more alive, personal, and immersive than text-based adaptation, which invited analytical scrutiny rather than emotional engagement. Players attributed intelligence and personal understanding to the preauthored system, coconstructing the perception of adaptive empathy through their own interpretive frames. A qualitative shift occurred between playthroughs, from exploratory orientation to strategic or emotional deepening. Emotional meaning was anchored in the narrative character rather than the system interface.

Discussion The findings suggest that designers of emotionally adaptive systems should embed adaptation in atmospheric and sensory channels rather than explicit narrative commentary, reduce the visibility of the system's adaptive voice, and design for interpretive ambiguity that allows players to coconstruct the experience of being understood. Limitations include the use of a single text-based research instrument and a small convenience sample. Future research should test the transferability of these findings across game genres and explore triangulation with physiological measures.

Acknowledgements

I would like to thank my supervisor, Donald McMillan, for his guidance and honest feedback throughout this project. I am grateful to the ten participants who gave their time and shared their experiences openly, without whom this study would not exist. I also want to thank the teachers and peers at DSV who shaped how I think about design and research over the past two years. Finally, I thank my family and friends for their patience and support while I built and wrote this.

Abbreviations

Abbreviation	Description
ACE	Assist, Challenge, and Emote (framework)
AI	Artificial Intelligence
API	Application Programming Interface
DSV	Department of Computer and Systems Sciences
ELARA	Emotional Landscape and Retrieval Architecture
IPA	Interpretive Phenomenological Analysis
MDA	Mechanics, Dynamics, and Aesthetics (framework)
PCG	Procedural Content Generation
PXI	Player Experience Inventory

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1

Introduction

Games have always been about feeling something. Whether through competitive tension, narrative immersion, or the quiet weight of a choice, digital games engage players emotionally in ways that distinguish them from passive media. Yet the emotional landscape of a game has historically been designed in advance, with fixed atmospheres, predetermined dialogue, and authored endings. The player's emotional state is inferred, at best, through difficulty curves and pacing. It is rarely treated as information that the game itself can respond to in kind.

In recent decades, a strand of research and design practice has emerged around emotionally adaptive games: systems that observe player behaviour and adjust their output in ways intended to sustain or deepen emotional engagement [12, 34]. These systems range from biofeedback driven difficulty adjustments to narrative branches conditioned on detected affect. The concept rests on a compelling premise, that a game can know something about how a player feels, and act accordingly.

Knowing something about how a player feels, however, is not the same as responding to it meaningfully. Two systematic reviews of the affect adaptive games literature reveal a consistent limitation: research overwhelmingly relies on physiological signals, survey instruments, and performance metrics to determine whether adaptation produces measurable effects [7, 29]. Across 162 affective gaming publications, Robinson et al. found that physiological sensing dominates, with insufficient critical reflection on what players actually experience [29]. Croissant et al. screened 3,930 papers and found not a single study in their 26-paper final sample that used in depth qualitative interviews as a primary method [7]. The field can say whether adaptation works. It cannot yet explain what being adapted to feels like.

The problem underlying this thesis is that game designers building emotionally adaptive systems currently lack qualitative understanding of how players subjectively experience adaptation. Adaptive systems can express responsiveness through different *adaptation modes*, distinguished by the type of game content that carries the adaptive response. Yannakakis and Togelius separate adaptable content into categories including narrative elements (dialogue, story branching) and presentational elements (visual dynamics, audio, camera) [34]. When a designer decides which adaptation mode to invest in, that decision shapes what players will feel, yet no study has investigated how different modes produce different experiential qualities. Without this understanding, designers are making consequential design choices based on technical feasibility and quantitative metrics (arousal levels, completion rates, self reported enjoyment) while remaining uninformed about the experiential texture each choice creates. This study aims to produce qualitative findings that inform those design choices directly, giving designers an evidence base grounded in how players actually experience different adaptation modes.

This thesis addresses that problem. It investigates player experience of emotionally adaptive game narratives through a qualitative lens, foregrounding the subjective and interpretive dimensions of the encounter between player and adaptive system. The study uses a purpose-built narrative game, *Fabricade*, as a research instrument. Fabricade is a text based interactive fiction game set in a near-future grief therapy context, in which an AI system called ELARA (Emotional Landscape and Retrieval Architecture) guides players through a sequence of constructed memories of a lost friend. The game implements two distinct conditions of emotional adaptation. In Condition A (Dialogue), adaptation manifests through changes in narrative dialogue and character response. In Condition B (Atmosphere), adaptation manifests through changes in atmospheric and aesthetic presentation, specifically vignette and glow overlays, text glitch effects, and screen glitch effects, while all narrative content remains identical. The choice structure and story events are the same across both conditions. Only the mode through which the system responds to player behaviour differs. Figure 1.1 shows a representative moment of play.

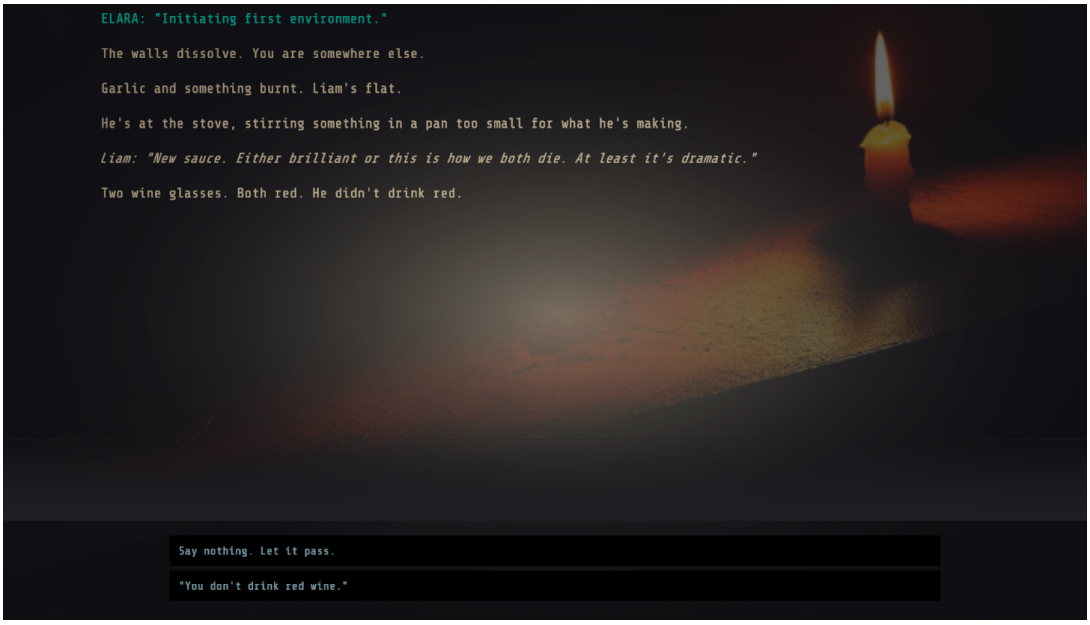


Figure 1.1: A representative moment of play in Fabricade (the dinner scene). ELARA's system narration appears in the accent colour, Liam's dialogue in italics, and the player selects from the choices at the foot of the screen.

The design of Fabricade is itself a form of argument. The game's central narrative disclosure, that the memories players have inhabited were algorithmically constructed by ELARA rather than faithfully retrieved, makes the logic of emotionally adaptive systems legible within the fiction. In doing so, the instrument foregrounds what adaptive systems typically conceal: the gap between manufactured responsiveness and felt meaning.

Ten participants each played both conditions in a counterbalanced within-subjects design. After each condition, participants completed a structured questionnaire. After both conditions, they participated in a semi-structured interview focused on specific moments of adaptive experience. Interviews were analysed using reflexive thematic analysis [4]. Automated session logs recorded player choices and hesitation times during play and provided behavioural data for triangulation. The full methodological rationale and procedure are detailed in Chapter 3.

1.1 Research Question

The research question guiding this thesis is:

> *How do players experience emotionally adaptive game narratives when the mode of adaptation varies?*

The question is exploratory and experiential: it seeks to describe and interpret subjective experience rather than measure causal effects or test hypotheses. The comparative element, examining two adaptation modes within the same game, is intended to reveal how the mode through which a system responds shapes the quality of the player's experience, not to determine which mode is superior. The study is scoped to narrative games that adapt emotional content, distinguishing it from the difficulty-based adaptation that dominates the existing literature. The intended contribution is qualitative design knowledge: insight into how different adaptation modes produce different experiential qualities, so that designers building emotionally adaptive systems can make informed decisions about where to invest adaptive effort.

1.2 Scope and Delimitations

This study focuses on emotional adaptation in narrative games and does not address adaptation in action, strategy, or other genres where emotional engagement is secondary to mechanical challenge. The adaptive behaviour in Fabricade is preauthored and rule based, and it does not use machine learning or real time biometric sensing, which permits controlled comparison between conditions without the variability of generative systems. The rationale for this choice is detailed in Chapter 3.

Fabricade is a purpose-built research instrument, not a commercially released product. Findings should be understood as contributions to qualitative design knowledge, not as evaluations of a system's technical accuracy. The study does not seek statistically generalisable results. Its contribution lies in the interpretive depth of participant accounts.

1.3 Thesis Overview

Chapter 2 reviews the literature on emotionally adaptive games, player experience theory, and the specific research gap this study addresses. Chapter 3 describes the research design, the Fabricade prototype, participants, data collection procedure, and analytical approach. Chapter 4 presents the qualitative findings. Chapter 5 discusses those findings in relation to the theoretical frameworks, addresses limitations, and considers implications for future research and design practice.

2

Extended Background

2.1 Affective Computing and Emotionally Adaptive Games

The concept of emotional adaptation in games originates in affective computing, defined by Picard as computation "that relates to, arises from, or deliberately influences emotions" [27]. Applied to games, this gives rise to systems that do not merely present emotional content but respond to the player's inferred emotional state. Gilleade, Dix, and Allanson formalised this idea through the Assist, Challenge, and Emote (ACE) framework, which identifies three adaptive strategies: the game can *assist* the player when frustration is detected, *challenge* the player when boredom is detected, or *emote* with the player by synchronising its aesthetic or narrative tone with the player's inferred state [12].

The Emote mode is the least studied of the three strategies. The majority of affect adaptive game research has operationalised adaptation through difficulty and challenge, leaving the emotional content dimension largely unexplored [7]. A 2023 systematic review confirmed that the field gravitates toward dynamic difficulty adjustment as its dominant form, with little work addressing stress, anxiety, or emotion driven adaptations in narrative contexts [7]. This study focuses specifically on the Emote mode, operationalised through two adaptation modes: narrative content (dialogue and story branching) and atmospheric presentation (visual overlays, text effects, and screen effects).

Yannakakis and Togelius extended affective game design through experience driven procedural content generation (PCG), which explicitly distinguishes content types that can be adapted, separating "stories, dialogue, quests" from "music, camera, visual dynamics" [34]. This taxonomy provides direct theoretical grounding for treating narrative and atmospheric adaptation as distinct adaptation modes. The distinction matters empirically: Croissant et al. found that no study has systematically isolated and compared these adaptation modes at the level of player experience [7].

Importantly, the studies reviewed above frame their contributions in terms that are directly relevant to game designers and developers. Gilleade et al. present the ACE framework as a design tool: the three adaptive strategies are offered as options a designer can choose between when structuring how a game responds to player affect [12]. Yannakakis and Togelius enumerate specific content types (narrative, audio, visual dynamics) that designers can target for adaptation, explicitly framing these as design variables rather than theoretical abstractions [34]. Croissant et al. call for "better methodological standards for game evaluation studies" and present their findings as guidance for researchers and practitioners designing adaptive systems [7]. In each case, the contribution is structured so that a designer reading the work can extract actionable implications for their own systems.

This study follows the same orientation. Its findings are organised by adaptation mode so that a designer can see what each mode (narrative content versus atmospheric presentation) produces experientially, what players notice, what they interpret, and what they feel. Where the findings suggest that one mode engages players differently from another, those differences are discussed as design implications: practical considerations for designers deciding where to invest adaptive effort in their own systems.

2.2 Generative and Large Language Model Approaches to Narrative Adaptation

The adaptation strategies described above are rule based: a designer specifies in advance what the system will do in response to a given player state. A parallel line of work pursues adaptation that is generated at runtime. Riedl and Bulitko survey interactive narrative as an intelligent systems problem, distinguishing systems that select among authored branches from those that construct narrative content dynamically, and note that the latter trades authorial control for responsiveness [28]. Large language models have intensified this trade. Värtinen et al. fine tuned GPT models to generate role playing game quest descriptions and quest giver dialogue, demonstrating that generated narrative content can reproduce the structural conventions of the form [32]. Park et al. extended this further with generative agents that store a record of their experiences, reflect on them, and plan subsequent behaviour, producing believable individual and collective conduct in a simulated environment [26].

These systems are relevant to the present study in two ways. First, they represent the direction in which emotionally adaptive narrative is now moving, which makes the question of how players experience adaptation more urgent rather than less. Second, they make the experimental problem harder. A generative system cannot hold the narrative constant across conditions, because its output varies with every run, and it cannot guarantee that two players encounter the same story events. Isolating the *mode* through which adaptation is expressed, which is the aim of this study, therefore requires authored content rather than generated content. The rule based instrument used here is not a weaker version of a generative system but a different tool chosen for experimental control, and the findings it produces concern how adaptation is experienced rather than how it is generated.

2.3 The Placebo Effect and Preprogrammed Adaptation

A significant body of evidence has demonstrated that perceived adaptation is experientially consequential, regardless of whether real time detection underlies it. Denisova and Cairns demonstrated the placebo effect in digital games: players told a game contained adaptive AI reported significantly higher immersion than controls, even when no such AI existed [9]. Their 2019 follow-up confirmed across two experiments (N = 180) that "the mere awareness of a digital game containing adaptive features is enough to significantly increase immersion," regardless of whether adaptation was actually implemented [8]. Kosch et al. extended these findings beyond games, showing in two experiments (N = 369 and N = 100) that the belief in receiving adaptive AI support increased performance expectations and sustained them after interaction [20].

These findings have direct methodological consequences for this study. Preprogrammed adaptation that genuinely responds to player choices (through rule based scripting) will be experienced as emotionally adaptive. The Barnum effect [11], the tendency to accept general descriptions as uniquely

applicable to oneself, and the ELIZA effect [33], the tendency to attribute understanding and intelligence to responsive computational systems, further support this. Players readily attribute emotional intelligence to systems exhibiting responsive behaviour, even when the underlying mechanism is a deterministic script. This means the study's choice to use preprogrammed adaptation rather than real time physiological sensing is not a compromise but a methodologically grounded decision that enables experimental control while preserving the ecological validity of the adaptive experience.

2.4 Interactive Fiction and the Poetics of Choice

Because the research instrument used here is a text based interactive fiction, the design tradition of that form is directly relevant. Montfort traces interactive fiction from the riddle through *Adventure* and the Infocom titles to the independent authoring communities that followed, and argues that the form's distinguishing feature is the coupling of literary text with a world model that responds to the reader [23]. What a work of interactive fiction offers the reader is not only prose but the experience of acting on it.

Mawhorter et al. make this concrete through choice poetics, a framework for analysing the narrative effect of a choice rather than its branching consequences [22]. In their account a choice has effects at the moment it is encountered, through the options presented and what they imply about the player character, regardless of what follows. This distinction matters for the present study. In Condition B (Atmosphere) the options are identical to those in Condition A (Dialogue), and the narrative outcome is held constant, yet the moment of choosing is framed by a different atmospheric context. Choice poetics gives a vocabulary for the possibility that the same choice can feel different without any change to its branches, which is precisely the distinction the two conditions are designed to isolate.

2.5 Player Experience in Narrative Games

Understanding player experience in narrative games requires frameworks attentive to both formal game structures and the subjective experience of play. The Mechanics, Dynamics, and Aesthetics (MDA) framework provides a foundational structure: game mechanics produce emergent dynamics, which in turn produce aesthetic responses (felt experience) in the player [14]. For adaptive games, this relationship is recursive: player behaviour shapes the mechanics, which reconfigure the dynamics, which alter the aesthetic, which shape subsequent behaviour. The player is simultaneously subject and coconstructor of the adaptation.

Ermi and Mäyrä proposed an experientially grounded model of immersion comprising three dimensions: *sensory immersion* (perceptual absorption in audiovisual stimulation), *challenge based immersion* (flow arising from balanced skill demand), and *imaginative immersion* (absorption in the narrative world and its characters) [10]. This tripartite model maps directly onto the two adaptation conditions under study. Condition A (Dialogue), which varies narrative dialogue and character response, primarily targets imaginative immersion: the player's investment in story and character is the medium through which adaptation operates. Condition B (Atmosphere), which varies atmospheric and aesthetic presentation through visual overlays and glitch effects, operates closer to sensory immersion, though in a text based game this manifests as changes to the affective quality of the reading experience rather than as the rich audiovisual absorption Ermi and Mäyrä originally describe. If the conditions produce different experiential qualities, Ermi and Mäyrä's dimensions offer vocabulary for explaining which perceptual pathway each condition engages.

Calleja argues that the core experience of playing is best characterised not as immersion but as *incorporation*: a reciprocal process in which the player internalises the game world while the game internalises the player's choices and agency [5]. In narrative games, incorporation is mediated by investment in characters and stakes in story outcomes. For adaptive systems, this raises a specific question: does a player's sense of incorporation deepen when the game responds to them, or does awareness of that responsiveness create a sense of being managed?

Isbister engages this question directly, observing that emotionally designed games risk producing effects that feel mechanical or manipulative at precisely the moment they succeed technically [17]. Players are sensitive to the difference between feeling genuinely moved and feeling deliberately triggered. Bopp et al. offer a complementary perspective, demonstrating that players actively construct meaning from emotionally challenging game experiences, engaging in interpretive work that extends beyond the session [1]. This finding supports the use of semi structured interviews focused on specific adaptive moments: participants are expected not merely to report felt states but to describe and reflect on what they experienced and how.

The Player Experience Inventory (PXI) identifies validated dimensions of player experience including immersion, mastery, autonomy, meaning, and curiosity [31]. These dimensions provide a useful vocabulary for structuring post gameplay questionnaires: items informed by PXI dimensions can serve as a supplementary measure that enables structured comparison across conditions and provides shared vocabulary that enriches a subsequent qualitative interview.

2.6 Grief Technologies and the Digital Afterlife

The scenario used in this study, an AI system that reconstructs a lost friend from a user's own account of him, is not purely fictional. Öhman and Floridi describe a digital afterlife industry in which the data of the deceased is reused commercially, and argue that such services require an ethical framework because the dead cannot consent and the bereaved are an unusually vulnerable customer [25]. Krueger and Osler examine conversational agents built from a dead person's messages, arguing that such systems can support what bereavement research calls continuing bonds, the maintenance of a relationship with the deceased rather than detachment from them, while also risking a dependence that forecloses adjustment [21]. Hollanek and Nowaczyk-Basińska extend this to generative systems, setting out design scenarios for griefbots and postmortem avatars and distinguishing the positions of the person whose data is used, the person who holds it, and the person who interacts with the result [13].

This literature informs the present study in two ways. It establishes that the premise of Fabricade describes a technology that already exists in commercial form, so participants were responding to a plausible system rather than to science fiction. It also sharpens the ethical reading of the findings. If players attribute understanding to a system that is only a deterministic script, as the placebo and ELIZA literature predicts, then the same attribution is available to a commercial griefbot, where the consequences of misattributed understanding fall on someone who is grieving. The design implications drawn in Chapter 5 are therefore stated with that application in view.

2.7 The Research Gap

Three convergent limitations characterise the existing literature.

First, qualitative inquiry is virtually absent. Croissant et al. found that across all 26 studies in their systematic review, evaluation methods consisted overwhelmingly of physiological signals, standardised questionnaires, and performance metrics, and not a single study used in depth qualitative interviews as a primary method [7]. Robinson et al., reviewing 162 affective gaming publications, similarly identified "insufficient critical reflection" on what players actually experience, noting that the field can measure arousal changes but cannot explain the interpretive and meaning making dimensions of adaptive play [29].

Second, mode comparison is absent. No published study has systematically compared how different adaptation modes shape player experience within the same game. Nogueira et al. adapted atmospheric elements in a horror game but did not isolate individual adaptation modes [24]. The theoretical frameworks distinguish these modes clearly. The ACE framework separates Emote from Assist and Challenge [12], Yannakakis and Togelius separate narrative from audiovisual content [34], and Ermi and Mäyrä separate imaginative from sensory immersion [10]. Yet empirical comparison between adaptation modes at the level of player experience has not been conducted.

Third, narrative adaptation specifically is understudied. The dominant paradigm remains difficulty adjustment, and story based adaptive systems remain at the research prototype stage with their experiential effects unevaluated through direct qualitative inquiry [34].

Croissant et al. called for "better methodological standards for game evaluation studies" [7]. Robinson et al. identified "insufficient critical reflection" in the field's methods [29]. This study addresses both calls directly, using a controlled research instrument to isolate adaptation modes and a qualitative methodology capable of producing the experiential depth that existing methods cannot generate. The intended outcome is design knowledge: qualitative accounts of how each adaptation mode shapes player experience, presented so that game designers building adaptive systems can make informed decisions about which adaptation modes to invest in and what experiential qualities those modes are likely to produce.

3

Method

3.1 Research Design and Methodological Approach

This study adopted a qualitative, interpretive research design situated within a constructivist epistemological framework. The central interest is in how players make sense of emotionally adaptive game narratives, an inherently interpretive question requiring methods capable of generating rich, participant centred accounts of experience. Following Braun and Clarke, the epistemological position treats participants' accounts as reflecting genuine experiences while acknowledging the researcher's active role in interpreting those accounts [4].

The study used a purpose built research instrument, the Fabricade prototype, as a technology probe [15]. Technology probes are simple, flexible instruments deployed in real use contexts to understand how people experience and respond to a design concept. The technology probe approach was appropriate here for three reasons. First, the research question asks how players experience adaptive narratives, not how to iteratively improve the design of an adaptive game. The probe exists to generate experiential data, not to evolve through design cycles. Second, the two condition structure of Fabricade requires a stable, controlled instrument: iterative redesign between participants would undermine the comparability of accounts across conditions. Third, Hutchinson et al. describe technology probes as instruments that "collect data about the use and the users" of a technology concept in context [15], which aligns precisely with this study's aim of collecting rich qualitative accounts of how players experience two modes of emotional adaptation.

Unlike Research through Design, which foregrounds iterative design and reflection on design choices as part of the knowledge contribution [35], the technology probe approach treats the artefact as a means of eliciting data about user experience rather than as a design outcome in itself. Unlike a lab experiment with controlled independent variables and statistical hypothesis testing, the technology probe is deployed in a naturalistic play context with qualitative rather than quantitative analysis as the primary mode of inquiry. Fabricade is not offered as a contribution to game design. It was deployed as a controlled instrument that exposed participants to two distinct modes of emotional adaptation and generated the experiential accounts that constitute this study's data.

The qualitative analysis of that data followed reflexive thematic analysis as described by Braun and Clarke [4, 2, 3]. Reflexive thematic analysis is an interpretive method in which themes are generated through the researcher's active engagement with the data, rather than discovered as pre existing patterns. This approach was chosen over two alternative qualitative analysis methods. Interpretive

Phenomenological Analysis (IPA) was considered but rejected: IPA requires a homogeneous sample with shared lived experience and focuses on the meaning of a particular phenomenon for a particular group [30]. This study's interest is in comparative experience across two conditions, not in the deep structure of a single phenomenon, and the convenience sampling approach does not select for the kind of homogeneous sample that IPA requires. Grounded theory was also considered but rejected because its aim is to generate a substantive theory grounded in data [6], which exceeds the scope of this study. The research question asks how players experience adaptation, not what theoretical model explains that experience. Reflexive thematic analysis is well suited to exploratory questions about subjective experience and produces findings that are accessible to and actionable for game designers, the intended audience for this study's contribution.

Several alternative study designs were also considered. A case study of an existing commercial adaptive game, such as a title using Valve's AI Director system, was rejected because it would introduce uncontrolled variables across narrative quality, production values, and player familiarity, making it impossible to attribute experiential differences to adaptation mode rather than extraneous factors. An experiment using physiological sensors for real time emotion detection was also considered but rejected on grounds of feasibility, ecological validity, and established evidence that preprogrammed adaptation that responds to player choices produces authentic adaptive experience [9, 8]. A purely quantitative survey based design was rejected because, as documented in Chapter 2, the field already has extensive quantitative data on adaptive game effects but lacks the qualitative depth needed to explain what adaptation feels like from the player's perspective [7, 29].

3.2 Research Instrument: Fabricade

Fabricade is a text based interactive fiction game developed in Unity using the Ink narrative scripting language [16]. Ink is an open source language designed for interactive narrative, used commercially in titles including 80 Days, Heaven's Vault, and Sea of Thieves. It tracks choices, variables, and visited content automatically, making it well suited for adaptive branching and behavioural logging. The ink unity integration plugin provides a C# runtime API for bidirectional communication between Ink story variables and Unity game objects.

The game's narrative places the player in the role of a person who has recently lost a close friend, Liam. The player enrols in a clinical programme run by ELARA (Emotional Landscape and Retrieval Architecture), an AI system that reconstructs shared memories as a form of grief therapy. Across six scenes (Intake, The Dinner, The Park Bench, The Hospital Corridor, The Room, and Discharge), the player makes one narrative choice per scene. These choices accumulate into a behavioural profile tracked by internal Ink variables (*openness*, *deflection*, *resistance*, *emotional posture*, *trust in the system*, and *mystery awareness*), which drive each condition's adaptive responses.

Designing the Narrative

The narrative was authored before either adaptive layer was implemented, so that both conditions could share an identical backbone. Three design requirements shaped it. First, the premise had to generate genuine emotional stakes, because a study of emotional adaptation requires that players have something to feel; grief was chosen for this reason, and because a system that reconstructs a lost person is itself an instance of the emotionally adaptive technology the study examines (Section 2.6). Second, the story had to accommodate adaptation without requiring it, so that Condition B (Atmosphere) could

hold every line of text constant and still be a complete narrative. Each scene was therefore written to be coherent on its own terms, with the adaptive layer added as variation rather than as load bearing plot.

Third, the choice points had to record a stance rather than a correct answer. Each of the six choices offers options that differ in how the player positions themselves toward ELARA and toward the memory, for example confronting an inconsistency, accepting it, or declining to answer, and no option advances the player further or rewards them. The six Ink variables were defined to capture that stance, which is why they track openness, deflection and resistance rather than progress or score. This follows the distinction drawn by choice poetics between the effect of a choice at the moment it is made and its branching consequences [22]: in this instrument the branching consequences are deliberately slight, and the experiential weight sits in the framing of the options.

The revelation that the memories are fabricated was placed in the fifth scene rather than the last, so that players had one further scene in which to respond to it. Session logs show this to be the point of longest hesitation in the game (Chapter 4), which indicates the placement functioned as intended.

Two conditions of adaptation were implemented:

Condition A (Dialogue): Narrative Dialogue Adaptation. The dialogue spoken by Liam and ELARA, and the player's internal monologue, adapt based on accumulated behavioural variables. A player who consistently confronts the system encounters substantively different narrative content than one who accepts it. Adaptation is expressed through *what* is said.

Condition B (Atmosphere): Atmospheric Aesthetic Adaptation. All narrative text and dialogue remains identical regardless of player choices. Instead, the game's atmospheric presentation adapts through three visual mechanisms: vignette and glow overlays that shift in intensity and warmth based on the player's accumulated emotional posture, text glitch effects in which characters momentarily scramble into symbolic noise at moments of narrative disruption, and a screen glitch effect triggered at the threshold between the hospital corridor and the revelation scene. ELARA's dialogue is also reduced to minimal, procedural responses, letting the atmosphere communicate rather than the system's voice. Adaptation is expressed through *how* the text feels to read, not what it says.

Both conditions share the same six scenes, the same choice points, and the same narrative backbone. They are distinguished solely by the layer through which adaptation is expressed: semantic content versus sensory atmosphere. This controlled distinction enables direct comparison of player experience across adaptation modes. Figure 3.1 illustrates the shared and distinct elements across both conditions, and Figure 3.2 shows the same narrative moment as it appears in each condition, holding the text constant while the atmospheric layer differs.

3.3 Participants

Participants were recruited through convenience sampling from the researcher's personal and academic network. The sample comprised 10 participants, a size consistent with qualitative research seeking experiential depth rather than statistical generalisation [3]. No prior gaming experience was required. However, participants needed to be comfortable reading English text at a steady pace, as Fabricade is entirely text based. No restrictions were placed on gaming background, and the sample included both gamers and non gamers. Beyond confirming student status and comfort with written

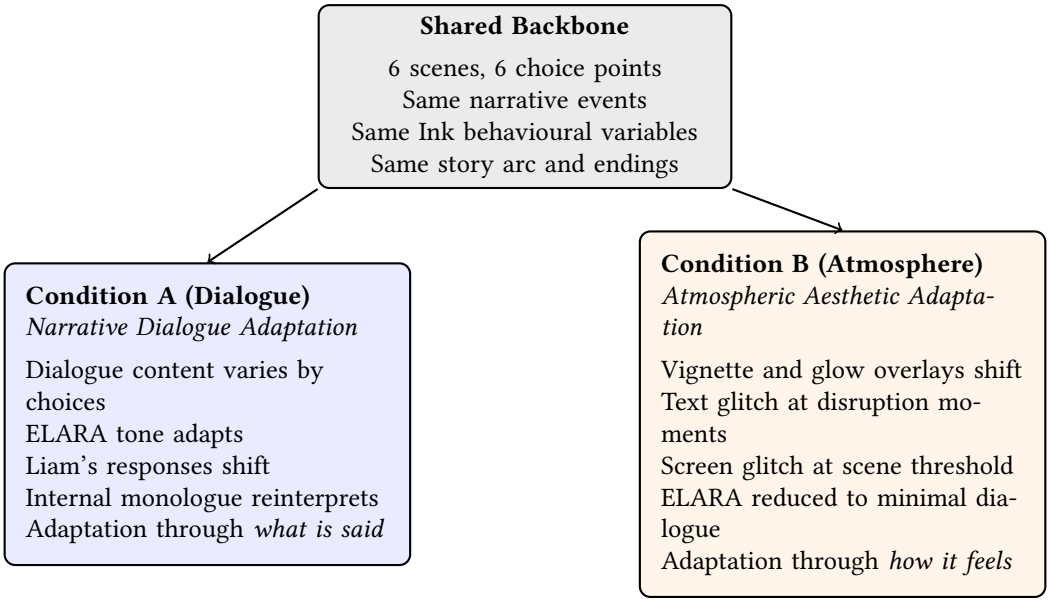


Figure 3.1: Shared and distinct elements across the two adaptation conditions in Fabricade.

English, no formal demographic data was collected. Where relevant, participants' own descriptions of their gaming background during interviews are used to contextualise their accounts.

Recruiting from a personal network carries a risk that extends beyond who was available to take part. Because participants knew the researcher had designed the game, there was a standing incentive for them to report positive experiences and to withhold criticism during the interview itself, which is the point at which the primary data was produced. Three measures were taken against this. Participants were told at the outset that the study examined how the game was experienced rather than whether it was good, and that accounts of confusion, boredom or disengagement were as useful as positive ones. Interview questions were phrased around specific moments rather than overall judgement, asking what a participant noticed or did at a given point rather than whether they enjoyed it, which makes an undifferentiated positive answer harder to give. And because each participant played both conditions, much of the analysis rests on comparisons internal to a single participant's account, where a general wish to be encouraging applies equally to both conditions and therefore does not explain a stated preference between them. These measures reduce the risk but do not remove it, and the limitation is revisited in Section 5.2.

3.4 Procedure

The study followed a within subjects design: each participant played both Condition A (Dialogue) and Condition B (Atmosphere) in a single session lasting approximately 75 minutes. Sessions were conducted remotely over Google Meet, with participants playing Fabricade in their web browser (deployed via itch.io) while sharing their screen. Condition order was counterbalanced to control for order effects. Half of the participants played Condition A (Dialogue) first, then Condition B (Atmosphere). The remaining half played in the reverse order. Participants were not informed of how the two conditions differed. They were told only that they would play two sessions of the same game.

Each session proceeded through five stages. First, the participant was briefed on the study and provided informed consent. Second, they played the first condition (approximately 5 to 15 minutes).

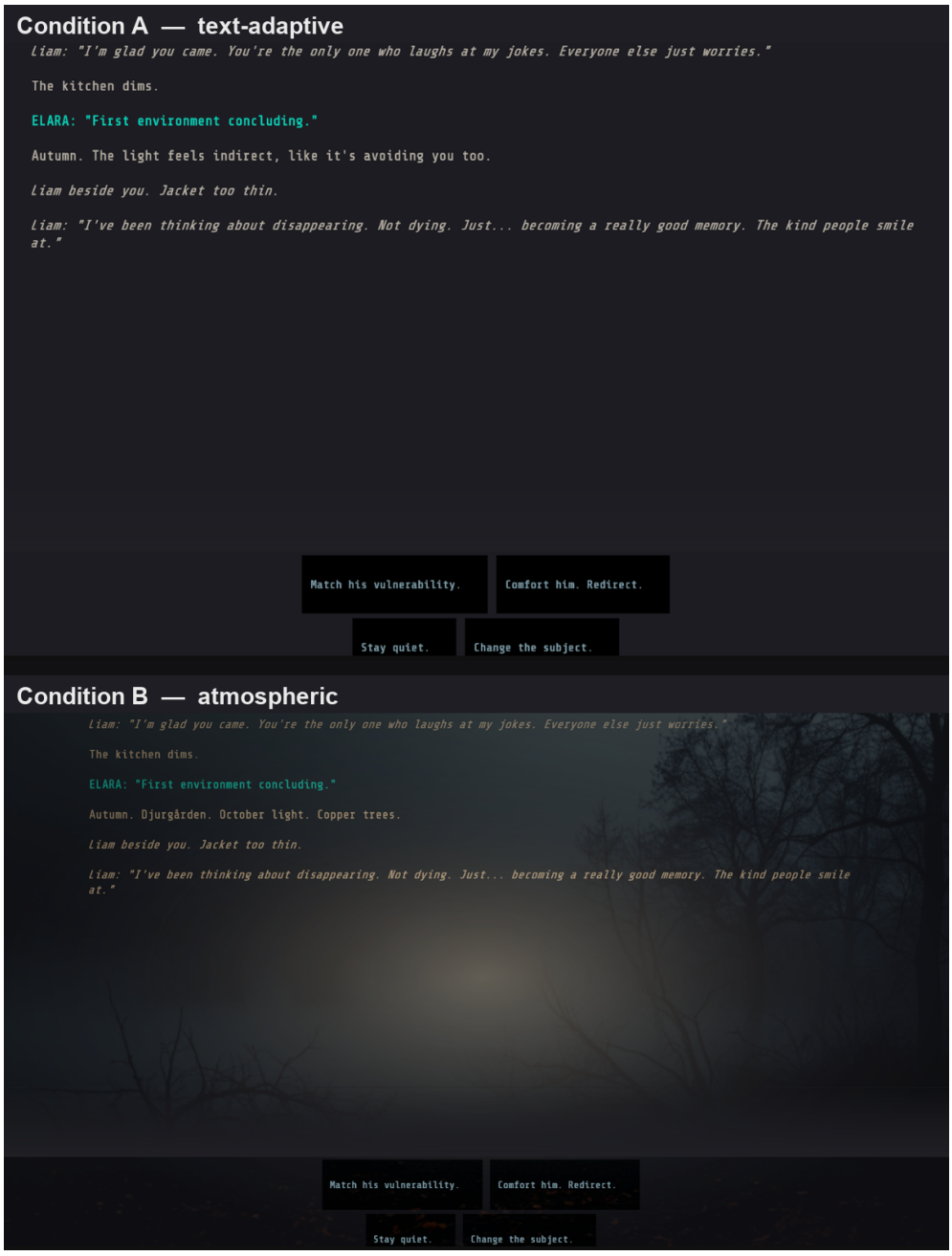


Figure 3.2: The same narrative moment (the park bench scene) rendered in each condition. In Condition A (Dialogue) (top), adaptation is carried by the dialogue and ELARA's commentary against a bare background. In Condition B (Atmosphere) (bottom), the narrative text is held constant and adaptation is carried by the atmospheric layer, here the autumn park setting behind the text. The choice options are identical across conditions.

Immediately after completing the first condition, they filled in a structured questionnaire. Third, they played the second condition and again filled in the questionnaire. Fourth, a semi structured interview was conducted (approximately 15 to 20 minutes). Finally, each session ended with a full debriefing in which the two conditions and the preprogrammed nature of the adaptation were explained, and participants were given a renewed opportunity to withdraw their data.

3.5 Data Collection

Data was collected through three instruments: a structured questionnaire, a semi structured interview, and automated session logs.

3.5.1 Questionnaire

After each condition, participants completed a short structured questionnaire (administered via Google Forms) consisting of nine items rated on a seven point Likert scale from "Strongly Disagree" to "Strongly Agree," followed by one open ended question. Six items measured player experience constructs informed by the dimensions identified in the Player Experience Inventory (PXI) [31]: control, meaning, attention, emotional involvement, immersion, and curiosity. Three additional items assessed perceived adaptiveness: whether the game felt responsive to the player's choices, whether the player's choices felt like they mattered, and whether the game seemed aware of how the player was engaging with it. The questionnaire provided a structured point of comparison across conditions and supplied shared vocabulary that enriched the subsequent interview.

3.5.2 Interview

The semi structured interview followed both conditions and focused on the experiential dimensions of each adaptive encounter. Questions were designed to elicit descriptions of *what* participants experienced and *how* they experienced it, rather than asking participants to explain *why* they felt a certain way. This orientation follows established qualitative practice for eliciting rich first person accounts [3]. The full interview guide is provided in Appendix B.

To support recall of specific moments, short video recordings of key adaptive events were prepared in advance for each condition: for example, the moment ELARA's dialogue tone shifted in response to a defiant choice in Condition A (Dialogue), or the moment text began glitching in Condition B (Atmosphere). These clips were shown to the participant during the interview as prompts, and questions were framed around them. Example questions included: "What was going through your mind when the text started glitching here?" and "How did it feel when ELARA responded differently to what you expected?" This approach grounded the interview in concrete moments of adaptive experience rather than abstract retrospection.

Interviews were audio recorded and transcribed for analysis.

3.5.3 Session Logs

Fabricade automatically logs all player interactions during each session. The session log records every scene transition, every choice presented, every choice made, and the hesitation time (in seconds) between a choice being presented and the player selecting an option. At the end of each session, the log captures the final values of all internal Ink variables (openness, deflection, resistance, emotional

posture, trust in the system, and mystery awareness) along with a labelled summary of choices made. Session logs provided behavioural data that was compared against participants' retrospective interview accounts, enabling triangulation between what players reported experiencing and what they actually did during play. The log structure and a summary of the closing state of all 20 sessions are provided in Appendix C.

3.6 Data Analysis

Transcribed interviews were analysed using reflexive thematic analysis following the six phase process described by Braun and Clarke [4, 2]. The process began with familiarisation through repeated reading of the transcripts, followed by systematic semantic and latent coding of segments relevant to the research question. Codes were then organised into candidate themes, which were reviewed against the coded data and the full dataset before being defined and named.

The analysis followed an inductive orientation: themes were generated from the data rather than mapped onto a predetermined framework. This is consistent with the study's exploratory intent and its aim of producing findings that are grounded in participant experience rather than imposed by theory. The researcher's positionality as the designer of Fabricade is acknowledged as shaping the analytical lens, enabling recognition of moments where player experience diverges from design intent while requiring vigilance against reading desired outcomes into participant accounts. Questionnaire data and session log data were used to contextualise and triangulate the qualitative findings, through descriptive summary and exploratory statistical testing, not as primary sources of evidence.

3.7 Ethical Considerations

Participants were provided with a written information sheet and gave informed consent before the session began (the consent form is provided in Appendix A). They were informed that participation was voluntary and that they could withdraw at any point without consequence. The game's narrative deals with themes of grief and loss. Participants were informed of this content in advance and reminded that they could stop the session at any time if they found the material distressing. Because participants were not told in advance how the two conditions differed, each session ended with a full debriefing in which the design of the study was disclosed and participants were offered a renewed opportunity to withdraw their data. None chose to do so. All data was stored securely and pseudonymised in reporting. Audio recordings were used solely for transcription and were not shared beyond the research team. The study followed the ethical guidelines of Stockholm University.

4

Results

Transcribed interviews from all ten participants were analysed using reflexive thematic analysis following Braun and Clarke [4, 2]. The analysis generated five themes that together address the research question: *How do players experience emotionally adaptive game narratives when the mode of adaptation varies?* Themes are presented below with supporting extracts from participant accounts. Questionnaire descriptives and session log data are integrated where they contextualise or triangulate the qualitative findings.

4.1 Participants and Sessions

Ten participants completed both conditions in a single session. Condition order was counterbalanced: five participants played Condition A (Dialogue) first (P1, P3, P5, P7, P9) and five played Condition B (Atmosphere) first (P2, P4, P6, P8, P10). Table 4.1 summarises session metadata drawn from the Fabricade session logs.

Participant	1st Condition	1st Duration	2nd Duration
P1	A	6.6 min	3.7 min
P2	B	8.6 min	9.6 min
P3	A	7.5 min	4.2 min
P4	B	7.3 min	7.5 min
P5	A	8.1 min	5.0 min
P6	B	5.7 min	5.6 min
P7	A	12.6 min	7.9 min
P8	B	4.2 min	6.8 min
P9	A	7.0 min	5.2 min
P10	B	4.6 min	6.7 min
Mean		*7.2 min*	*6.2 min*

Table 4.1: Session durations per participant. First playthroughs were generally longer (mean 7.2 vs. 6.2 minutes), with P7's first session the longest at 12.6 minutes.

Table 4.2 presents condition-level results from the post-play questionnaire (seven-point Likert scale). Because every participant completed both conditions, each item was compared between conditions using a Wilcoxon signed-rank test, the paired, non-parametric test appropriate for ordinal Likert

data, computed in JASP [18]. Given the small sample ($N = 10$), these tests are underpowered, and a non-significant result cannot be interpreted as evidence that no difference exists. For this reason the matched rank-biserial correlation (r) is reported alongside each p-value as an effect-size estimate that does not depend on sample size, and the questionnaire is used primarily to contextualise and triangulate the qualitative findings rather than to establish population-level effects. Consistent with the principle that at low N the direction and magnitude of median shifts remain interpretively useful even where statistical significance is not reached, both are considered below.

Item	A Mean	A Mdn	B Mean	B Mdn	W	p	r
Control	4.40	4.5	4.50	4.0	13.5	1.000	+0.04
Meaning	4.10	3.5	4.60	5.5	4.0	.410	+0.47
Attention	5.60	6.5	5.50	6.5	9.0	.766	−0.20
Emotional Involvement	3.50	3.0	4.30	5.5	7.0	.124	+0.61
Immersion	4.80	5.0	5.40	5.5	3.5	.071	+0.75
Curiosity	6.20	6.5	5.70	6.0	11.5	.341	−0.53
Responsiveness	4.40	4.5	4.20	4.0	6.5	.713	−0.30
Choice Weight	4.89*	5.0	4.70	4.5	11.5	.915	−0.10
System Awareness	5.30	5.5	6.10	7.0	0.0	**.031**	+1.00

Table 4.2: Condition-level questionnaire results ($N = 10$). Items 1–6 informed by PXI dimensions [31]. Items 7–9 assess perceived adaptiveness. W, p, and the matched rank-biserial correlation r are from Wilcoxon signed-rank tests comparing conditions. Positive r indicates higher ratings in Condition B (Atmosphere). Only System Awareness reached significance ($p = .031$), shown in bold. *$n = 9$ for Choice Weight in Condition A (Dialogue) (one participant left the item blank).

Only System Awareness differed significantly between conditions ($W = 0.0$, $p = .031$, $r = +1.00$): every participant who distinguished the conditions rated the atmospheric Condition B (Atmosphere) as more aware of them than the text-adaptive Condition A (Dialogue). Two further items showed large effects in the same direction but did not reach significance at this sample size: Immersion ($r = +0.75$, $p = .071$) and Emotional Involvement ($r = +0.61$, $p = .124$). Curiosity leaned the other way, favouring Condition A (Dialogue) with a moderate effect ($r = -0.53$, $p = .341$). Read with the caveat that $N = 10$ leaves little power to detect effects of this size, the pattern is consistent throughout: the atmospheric condition produced stronger felt involvement and a stronger sense of being understood, while the text condition produced marginally greater curiosity. These directions align with the qualitative themes that follow, which is the sense in which the questionnaire is treated as triangulating evidence rather than confirmatory testing.

Beyond session duration, the Fabricade logs recorded the hesitation time at every choice point, defined as the interval between a choice being presented and selected, used here as a behavioural index of deliberation. Table 4.3 reports mean hesitation per narrative choice point by condition. Two patterns recur and are taken up in the thematic analysis: the room reveal drew by far the longest deliberation, and hesitation was higher in the text-adaptive Condition A (Dialogue) than the atmospheric Condition B (Atmosphere) at every choice point except the opening one, which precedes any condition-specific content. A Wilcoxon signed-rank test confirmed the condition difference at the pivotal room reveal (Condition A (Dialogue) mean 83.2 s versus Condition B (Atmosphere) mean 53.2 s, $W = 1.0$, $p = .008$), indicating that making adaptation explicit through altered text and system commentary prompted measurably more deliberation than embedding it in atmosphere.

Choice point	All	Dialogue (A)	Atmosphere (B)
Room: response to the reveal	68.2	83.2	53.2
Corridor: "Are you real?"	40.4	45.4	35.4
Discharge: what to do with the data	35.2	43.3	27.1
Dinner: the wine slip	34.5	39.3	29.7
Bench: how to respond	32.0	34.8	29.2
Intake: reason for coming	24.0	24.0	24.0
Intake: favourite memory	20.1	21.3	18.8

Table 4.3: Mean hesitation (seconds) per narrative choice point, overall and by condition ($N = 10$ per condition). Hesitation is the time between a choice being presented and selected. Single-option "Continue" prompts, which carry no decision, are omitted.

A final use of the session logs was to check whether the behavioural profile each player accumulated through their choices, captured at the end of each session as the values of the internal Ink variables, corresponded to what players reported on the questionnaire. This check is exploratory: with twenty sessions pooled across ten participants and two conditions, without independence between a participant's two sessions, and across many variable pairs, the correlations are indicative rather than confirmatory. Spearman correlations nonetheless revealed a coherent pattern. The *mystery awareness* variable, which accumulates as a player registers the game's central conceit, was positively associated with self-reported Curiosity ($\rho = .66$, $p = .002$), the sense that choices mattered ($\rho = .72$, $p = .001$), perceived System Awareness ($\rho = .57$, $p = .009$), and Emotional Involvement ($\rho = .52$, $p = .018$). The *resistance* variable, which accumulates when a player pushes back against the system, was associated with self-reported Control ($\rho = .51$), Attention ($\rho = .50$), and Immersion ($\rho = .50$, all $p < .05$). The remaining variables (openness, deflection, emotional posture, and trust in the system) showed no significant associations. Both significant patterns are consistent with the qualitative account: players who engaged actively with the game's mystery, whether by pursuing it or resisting it, both registered that engagement behaviourally and reported it experientially. These correspondences are modest and should not be over-read, but they indicate that the behavioural log and the self-report were capturing related aspects of the same encounter, which supports their use together as triangulating evidence in the thematic analysis that follows.

4.2 Theme 1: "The Feeling Was Real"

The first and most overarching theme concerns emotional authenticity within fabricated experience. Players who invested in the narrative found the experience genuinely affecting. The knowledge that the game, the memories, and Liam were not real did not diminish the emotional response and, for some, actively intensified it. Seven of ten participants contributed to this theme.

P4, who played Condition B (Atmosphere) first, described the emotional conflict at the centre of the game:

> "On one hand, I was very much torn apart that I have to let this go, this is not real at all. On the other hand, I was wishing that I could experience it all, that I could be reincarnated so that I could meet Liam, experience whatever we had experienced while Liam was with me before he left me for good."

P4's account articulates a tension between cognitive awareness of fabrication and emotional investment in the experience. This tension did not resolve into detachment. It persisted and deepened

across the session. In a later response, he described the game as making Liam a "placeholder for a lot of people," linking the fabricated experience to personal losses: "during my lifetime I lost a lot of people very close to my heart."

P9 described a similar pattern from a different angle. When asked whether the fabricated nature of the memories affected his experience, he responded: "I was pretty immersed in the game. I felt it pretty well. I don't think it would make much of a difference to me even if it wasn't real." Where P4 was torn, P9's engagement simply overrode the distinction between real and fabricated. P8 arrived at a comparable position through more tentative language: "Just because it's not real doesn't mean how we act upon it doesn't matter or something like that. Maybe." P10 stated it most directly: "It doesn't really matter if it's memory or not, real or not."

P7, who described himself as not emotionally affected, nonetheless offered the most precise articulation of the theme's underlying logic:

> "The feeling was mine, the feeling was from the participant, and the fabrication was by the machine."

This formulation separates ownership of feeling from authorship of stimulus. The fabrication belongs to the system. The feeling belongs to the player. P3 engaged with a similar question from within the narrative itself, asking "whether it's the grief that is basically talking or taking over the memory, or is it the memory that is taking over the grief." P5 expressed emotional avoidance after the session, stating "if it takes me through the entire experience again, I don't want to involve myself in that," suggesting the experience carried weight beyond the session.

Session log data reinforces this finding. The room scene, where the game reveals that the memories are fabricated and asks players to respond, produced the longest mean hesitation of any choice point in the game (68.2 seconds, compared to 40.4 seconds for the next-longest choice point, the corridor "Are you real?" decision). A Friedman test across the seven decision points confirmed that hesitation differed significantly by choice point ($\chi^2(6) = 47.66$, $p < .001$), and the room reveal ranked as the longest mean hesitation for every participant. At the level of individual sessions it was the longest choice in 18 of 20; in the two exceptions (P3 and P6, both in Condition B (Atmosphere)) the corridor choice was longer by under six seconds. It was significantly longer than the next-longest choice point (Wilcoxon signed-rank $W = 0.0$, $p = .006$, $r = +1.00$). Of the twenty room scene responses across all participants and conditions, seven chose "It doesn't matter. I felt it" (acceptance), seven chose "I don't know what I feel anymore" (ambivalence), four chose "What about you? Do you feel this?" (turning the question back to the system), and only two chose "Then none of this counts" (rejection). The near-absence of outright rejection aligns with the interview finding: players overwhelmingly treated the feeling as real, even when the stimulus was not.

Three participants (P7, P8, and P6) were not emotionally affected. All cited not knowing Liam as the primary reason. P7 observed that the game "can be designed with somebody doing it for their loved ones" and would then "emotionally impact you." This boundary condition suggests the design's emotional power depends on whether the player invests in the fictional relationship, not on whether they recognise it as fictional.

4.3 Theme 2: "Show, Don't Tell"

The second theme concerns the comparative experience of the two adaptation conditions. Condition B (Atmosphere: images, sound, glitch effects, reduced ELARA commentary) was consistently perceived

as more alive, personal, and engaging than Condition A (Dialogue: altered dialogue, expanded monologue, ELARA commentary). Five of ten participants explicitly described Condition B (Atmosphere) as more alive. The remaining accounts provide nuance and counter-evidence.

P3 described a shift in perspective between conditions:

> "In Condition A, I felt like a third person that was observing those people's interaction. In Condition B, I felt like I am the other person that is trying to decipher what Liam was doing."

The distinction is between observation and participation. Condition A (Dialogue), despite containing more adaptive content in its dialogue, positioned P3 as an observer. Condition B, through atmospheric changes, drew her in as a participant. P3 attributed this partly to sensory detail: "the clattering of the knives and forks and the background images created more ambience."

P5 expressed the difference in terms of perceived humanity: "It kind of felt like a human talking to a human in Condition B." P10 framed it as the difference between play and evaluation: "Condition B felt more like a game where I am exploring and was actually having fun. Condition A felt more like a test phase where the system is shaking and trying to get to the right answer." P1, whose questionnaire scores were near-ceiling for Condition B (Atmosphere) (seven on eight of nine items), attributed the difference to imagination: "I could imagine it more."

The Glitch Divide

The visual glitch effect in Condition B (Atmosphere) (Figure 4.1) functioned as a dividing line between engaged and disengaged participants. P10 described the glitch as aesthetically meaningful:

> "There were some glitches in Condition B, which I'm not sure if by design or not. I liked it. It felt fractured. That was nice. [...] I felt like the system is hesitating to tell me something and they changed their answer. I liked that too. That felt, for lack of better words, alive."

P9 interpreted the same effect as an "intimidation tactic." P1 read the glitch as a character trait of ELARA. All three treated the effect as intentional and meaningful. In contrast, P6 interpreted the glitch as a system error: "I was thinking it's part of the game, and the glitch happened because ELARA is a system and something went wrong with that system." P7 found it "mostly like an annoyance." The same design element succeeded or failed based on the player's investment level.

Three participants found Condition A (Dialogue) more alive: P6, P7, and P8. For P6 and P8, Condition A (Dialogue) was their second playthrough, suggesting a possible familiarity effect. However, P7 played Condition A (Dialogue) first and still preferred it ("maybe Condition A, because it was unknown to me"), meaning preference for text adaptation is not fully explained by order effects. P2, who played Condition B (Atmosphere) first, completely missed its visual differences, suggesting atmospheric adaptation requires a baseline level of attention to register.

The questionnaire data provides quantitative context. Condition B (Atmosphere) scored higher on Emotional Involvement (mean +0.80, median +2.5), Immersion (mean +0.60), and System Awareness (mean +0.80). Condition A (Dialogue) scored slightly higher on Curiosity (mean +0.50). The atmospheric condition produced stronger felt experience. The text condition produced slightly more intellectual engagement.

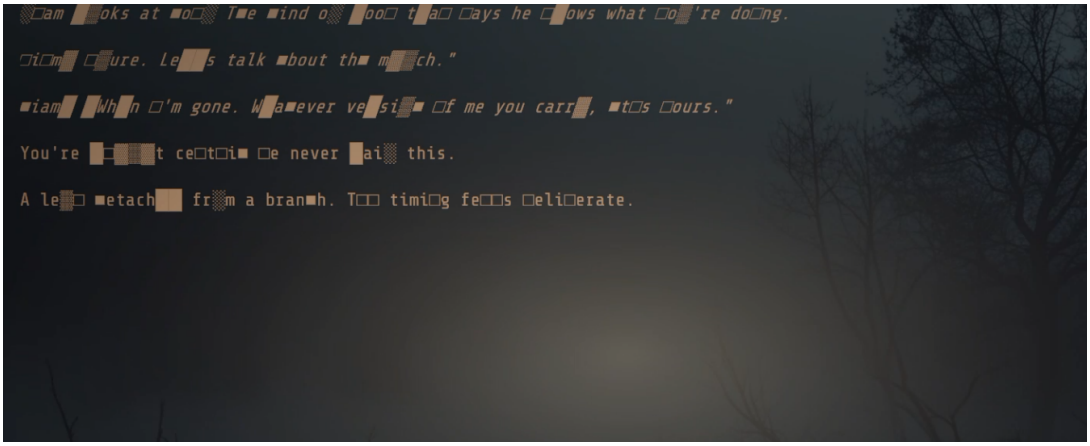


Figure 4.1: The text glitch effect in Condition B (Atmosphere), in which characters scramble into symbolic noise at a moment of narrative disruption. Some participants read the effect as intentional and meaningful. Others read it as a system malfunction.

4.4 Theme 3: "No One Has Asked Me This Before"

The third theme concerns perceived system intelligence. Despite the game being entirely preauthored with no AI, players consistently attributed intelligence, personal knowledge, and intentional responsiveness to it. The perception of adaptive intelligence was constructed by the player, not delivered by the system. Six of ten participants contributed to this theme.

P9 described a moment where the game appeared to recognise his strategy:

> "I thought that it really understood my strategy. What I was trying to do was withhold as much information from my side as I could and try to figure out as much as I could from its side. That was my intention and I think it observed it really well."

The system did not observe P9's strategy. The Ink scripting engine tracked his choices through six numerical variables. The response he experienced was a prewritten branch triggered by his accumulated resistance score. Yet the experience of being understood was genuine.

P5 described the system as "trying to know me personally" and identified a specific moment of perceived reciprocity. After choosing to ask Liam "What do you feel?" (turning the system's question back on itself), the game responded with the line "No one has asked me this before." P5 described this as a pivotal exchange: "I flipped the mirror for the system as well. It replied with 'No one has asked me this before.' So I really wanted to know what it meant." The line is authored dialogue, not a dynamically generated response, but P5 experienced it as the system genuinely registering his unexpected move. In a separate moment, when Liam responded "Yeah, that's what I thought" after P5 chose silence, P5 described the response as "kind of like an intimidating thing for me," a prewritten acknowledgement read as personal confrontation.

P4 described the game's effect in terms of psychological manipulation: "I feel like the game did some sort of cyber psychology with me that even though I didn't want control over the situation, it made me feel like I was having more control." P7, despite his analytical orientation, described satisfaction at communicating his engagement: "I felt good. I felt like I have successfully told the machine that I want to engage."

P10 experienced the inverse pattern. Rather than feeling understood, he felt misread: "I felt like the game thought I am not consistent, which is a bit odd because I tried to be." The session logs show that

P10 chose identically on four of seven choice points across conditions, but varied on three, enough for the system's variables to register inconsistency. His frustration confirms the broader finding: players expected the system to understand them, and the violation of that expectation was itself emotionally consequential.

P8 represented a boundary case. His gaming experience made him sceptical: "I'm not sure if the choices matter. Because I've played games before where you have alternate dialogues but they kind of lead to the same flow anyway." Gaming literacy can immunise against the illusion of system intelligence.

4.5 Theme 4: "Exploring, Then Solving"

The fourth theme concerns the qualitative shift between playthroughs. All ten participants described an evolution from their first to their second playthrough that went beyond mere familiarity. The first playthrough was characterised by orientation, confusion, and intuitive choice-making. The second brought strategic exploration, emotional deepening, or deliberate testing.

P3 named the shift directly: "On the first run, I was trying to just explore the mystery, but on the second run, I was trying to solve the mystery." Critically, she identified the shift as partially independent of the conditions themselves: "I believe that's not entirely contributed to Condition A or B. That can also be contributed because I have played through once, so my thinking process has been more matured."

P5 described the shift in emotional rather than strategic terms. In his first playthrough (Condition A (Dialogue)), choices were oriented toward understanding the system. In his second (Condition B (Atmosphere)), "the choices were real. The choices were based on pure emotion. I'm not saying there's no emotion in Condition A, but you can know the difference." The session logs confirm this: P5's room scene choice shifted from "What about you? Do you feel this?" (engaging with the system) in Condition A (Dialogue) to "I don't know what I feel anymore" (emotional overwhelm) in Condition B (Atmosphere).

P4 described a transformation in self-awareness:

> "The self-awareness was extremely strong from my side in Condition A [his second playthrough], whereas it was non-existent in Condition B [his first playthrough]."

His first playthrough was immersive and unguarded. His second was marked by the knowledge that Liam was not real and that the experience would end. The session logs show P4 changed every choice between conditions (zero of seven choices the same), shifting from open, confrontational responses in Condition B (Atmosphere) to avoidant, resistant ones in Condition A (Dialogue).

The session logs reveal distinct strategic profiles across participants. P9 changed all seven choices between conditions, confirming his interview statement: "I really wanted to figure out all the outcomes." P8 kept six of seven choices identical, confirming: "I picked the same answers for most of them on purpose." His only change was the room scene choice, where his hesitation more than doubled from the first to the second playthrough (34.5 to 86.8 seconds), suggesting the question accumulated weight through repetition. P2 described his first playthrough as "mostly a learning phase for me because without any context the game started."

The questionnaire captures the shift in individual scores. P9 rated Control at 1 for Condition A (Dialogue) (his first playthrough) but 5 for Condition B (Atmosphere), consistent with his account of first-playthrough disorientation giving way to second-playthrough agency.

4.6 Theme 5: "ELARA the Robot, Liam the Human"

The fifth theme concerns the experiential distinction between ELARA (the system interface character) and Liam (the narrative character). Nine of ten participants commented on ELARA. Players consistently attributed mechanical, procedural qualities to ELARA and human, emotionally resonant qualities to Liam. Emotional engagement was anchored in Liam's dialogue, his self-awareness, and his vulnerability, not in ELARA's adaptive commentary.

P5 stated the distinction most clearly:

> "ELARA really felt like a robot to me. But Liam [...] it really feels like a human being for a moment. And that's what influenced my choices from Condition A to Condition B."

P7 described ELARA as "unnecessary," adding: "I felt like it would be better, more immersive if ELARA was not there." P10 characterised ELARA as "a generic program that's been trying to communicate with me." P6 understood ELARA in functional terms: a system that could malfunction. These descriptions cluster around a common perception: ELARA is infrastructure, not character.

Liam, by contrast, was experienced as present and alive. P4 described ELARA as "somewhat half-self-conscious [...] not fully artificial intelligence for me, but it was not far behind," while describing Liam's moment of self-awareness, the point where Liam's dialogue reveals that he knows he is not real, as the moment that "stood out" most across both sessions. P4's engagement with Liam extended to protective behaviour: "I had to hide my face in a way to not let Liam know that I already knew he would not be with me in the next few moments." The player was shielding a fictional character from knowledge that the player already possessed, a behaviour that presupposes genuine emotional investment.

P9 engaged with ELARA differently, interpreting the system as adversarial: "If I'm being paranoid, I'd say the system was designed to intimidate or extract information." This reading positioned ELARA as an antagonist rather than a facilitator, and the glitch effects in Condition B (Atmosphere) reinforced this perception: "I thought it was an intimidation tactic, and the background as well." P3 offered a more clinical reading, understanding ELARA as "a memory reconstruction interface that basically helps people to deal with grief."

The distinction between ELARA and Liam maps onto the condition comparison. When ELARA was more present (Condition A (Dialogue)), participants described the experience as more structured. P8 observed: "It does feel like in Condition A that maybe it was reacting more strongly to my choices." When ELARA receded (Condition B (Atmosphere)), the experience felt more human and alive. The questionnaire data supports this: System Awareness was the highest-scoring item for Condition B (Atmosphere) (median 7.0 versus 5.5 for Condition A (Dialogue)). Players perceived the atmospheric condition as more aware of them despite it containing less explicit adaptive content, suggesting that reducing the system's voice paradoxically increased the sense of system presence.

4.7 Summary of Findings

Table 4.4 summarises the five themes and their contribution to the research question.

Across all five themes, one consistent boundary condition emerged: the depth of a player's engagement with the themes depended on whether they invested in the fictional relationship with Liam. Participants who did not invest (notably P7, P6, and P8) still engaged with the game's mechanics and

Theme	Core finding	Participants
1. “The Feeling Was Real”	Fabricated experiences produced authentic emotional responses. Knowledge of fabrication did not diminish engagement	7/10
2. “Show, Don’t Tell”	Atmospheric adaptation was perceived as more alive and personal than text-based adaptation	8/10
3. “No One Has Asked Me This”	Players attributed intelligence and personal understanding to a preauthored system	6/10
4. “Exploring, Then Solving”	Players underwent a qualitative shift between playthroughs, from orientation to strategic or emotional deepening	10/10
5. “ELARA the Robot, Liam the Human”	Emotional meaning was anchored in the narrative character, not the system interface	9/10

Table 4.4: Summary of thematic findings.

observed its adaptive behaviour, but the emotional weight that characterises Themes 1, 2, and 5 was absent from their accounts. This is not a separate finding but a moderating factor that conditions all five themes.

5

Discussion

This chapter interprets the findings presented in Chapter 4 in relation to the theoretical frameworks reviewed in Chapter 2, addresses the study's limitations, considers implications for game design practice and their ethical and societal weight, and concludes with an explicit partial answer to the research question together with directions for future work.

5.1 Interpretation of Findings

The research question asked how players experience emotionally adaptive game narratives when the mode of adaptation varies. The five themes identified through reflexive thematic analysis offer a composite answer: players experience adaptation not as a technical feature but as an interpretive encounter, in which the mode of adaptation shapes the quality of that encounter in distinct ways.

5.1.1 Atmospheric Adaptation and the Immersion Pathway

The finding that atmospheric adaptation (Condition B (Atmosphere)) was consistently perceived as more alive, personal, and engaging than text-based adaptation (Condition A (Dialogue)) maps onto Ermi and Mäyrä's distinction between sensory and imaginative immersion [10]. However, the mapping is not straightforward. Condition B (Atmosphere) was designed to target sensory immersion through visual overlays, glitch effects, and ambient sound. Yet participants described its effects in terms closer to imaginative immersion: P3 shifted from observer to participant, P5 felt a "human talking to a human," and P1 could "imagine it more." The atmospheric mode did not merely create sensory atmosphere. It deepened narrative engagement.

This finding complicates the clean separation between content types proposed by Yannakakis and Togelius, who distinguish "stories, dialogue, quests" from "music, camera, visual dynamics" as independent adaptation targets [34]. The present study suggests these content types are experientially entangled: atmospheric changes altered how players interpreted narrative content, even when that content was identical across conditions. Condition A (Dialogue), despite containing more semantically adaptive material, was perceived as more structured and evaluative. This aligns with Isbister's warning that emotionally designed games risk producing effects that "feel mechanical or manipulative at precisely the moment they succeed technically" [17]. Making adaptation visible through text invited analytical scrutiny. Embedding it in atmosphere allowed players to feel rather than assess.

The questionnaire data supports this interpretation. Condition B (Atmosphere) scored higher on Emotional Involvement and Immersion while Condition A (Dialogue) scored slightly higher on Curiosity, suggesting the two modes engage distinct experiential registers: atmospheric adaptation produces felt involvement, while text adaptation produces intellectual engagement.

5.1.2 Perceived Intelligence and the Coconstruction of Adaptation

Theme 3 ("No One Has Asked Me This Before") demonstrates that players attributed intelligence, personal understanding, and strategic awareness to a system that was entirely preauthored. This finding extends the placebo effect in games documented by Denisova and Cairns [9, 8]. Where Denisova and Cairns showed that *believing* a game is adaptive increases immersion even when no adaptation exists, the present study shows that when adaptation *does* exist, players attribute substantially more intelligence to the system than its mechanism warrants. P9 believed the system "really understood my strategy." P5 experienced a prewritten dialogue line as a personally directed confrontation, and P4 described the game as performing "cyber psychology." These attributions are consistent with the ELIZA effect [33] and the Barnum effect [11], but they operated here on a system that genuinely responded to player choices through rule-based scripting, not on a system that merely appeared responsive.

This finding has theoretical implications for how adaptation is understood within the MDA framework [14]. The aesthetic response (felt experience of being understood) was not a direct product of the mechanic (Ink variable tracking) or the dynamic (branching dialogue). It was cocreated by the player's interpretive frame. Players filled in gaps, attributed intentionality to coincidences, and experienced designed ambiguity as personal responsiveness. The adaptation was algorithmic. The empathy was constructed.

5.1.3 Emotional Authenticity and Meaning Construction

Theme 1 ("The Feeling Was Real") resonates with Bopp et al.'s finding that players actively construct meaning from emotionally challenging game experiences [1]. P4 projected personal losses onto Liam, P5 avoided re-engaging with the experience after the session, and P7 articulated the feeling-fabrication split with precision: "the feeling was mine [...] and the fabrication was by the machine." These are not passive responses to stimuli. They are acts of interpretation and meaning-making.

The boundary condition identified across all five themes, that emotional depth depended on whether the player invested in the fictional relationship with Liam, connects to Calleja's concept of incorporation [5]. Participants who incorporated (internalising the game world and having their choices internalised by it) experienced the themes at full depth. Those who did not invest in Liam engaged with the game's mechanics but not its emotional substance. This suggests that adaptive systems, however sophisticated, cannot manufacture incorporation. They can only respond to it once it occurs.

Theme 5 ("ELARA the Robot, Liam the Human") further illuminates this dynamic. Reducing the system's voice (ELARA) paradoxically increased the sense of system presence, as measured by the System Awareness questionnaire item (median 7.0 in Condition B (Atmosphere) versus 5.5 in Condition A (Dialogue)). When ELARA narrated the adaptation explicitly, players experienced it as the system *talking about* the experience. When ELARA receded, the adaptation was experienced as *being* the experience. This has clear resonance with the Emote mode of the ACE framework [12]: effective emotional adaptation may require the system to be felt rather than heard.

5.2 Limitations

Several limitations qualify the findings.

First, the study used a single research instrument. Fabricade is a text-based interactive fiction game with a specific narrative premise (grief therapy). Findings may not transfer to visually rich, mechanically complex, or tonally different games. The atmospheric adaptation in Condition B (Atmosphere) operated through visual overlays and text effects, not through the rich audiovisual environments available in commercial productions.

Second, the sample was small ($N = 10$) and drawn through convenience sampling from the researcher's personal and academic network. While sample size is appropriate for reflexive thematic analysis seeking experiential depth rather than statistical generalisation [3], the accounts reflect a specific demographic context. P7's observation that the game would be more impactful with real personal data suggests the artificial framing may have constrained the emotional range of responses.

Third, the within-subjects design means that playthrough order and condition are partially confounded. Theme 4 ("Exploring, Then Solving") documents a qualitative shift between playthroughs that occurred regardless of condition order, and participants such as P3 explicitly acknowledged the order effect. The counterbalanced design mitigates but does not eliminate this confound. Any comparison between conditions must be read alongside the evolution that repeated exposure produces.

Fourth, the researcher's dual role as designer of Fabricade and analyst of the data introduces a positionality concern. Familiarity with the design intent may have shaped which patterns were noticed and how they were interpreted. The reflexive orientation of the thematic analysis [2] treats this as a condition of qualitative inquiry rather than a methodological flaw, but readers should note that the analytical lens was shaped by proximity to the design.

Fifth, participants were recruited from the researcher's personal and academic network and were in most cases known to the researcher, who also conducted the interviews. In a study of emotional response this prior relationship is a potential source of bias: existing rapport may have encouraged candour, but social desirability may equally have led participants to soften criticism or amplify their reported engagement. The risk is sharpest during the interviews themselves, which are the point at which the primary data was produced, and it is compounded by participants knowing that the researcher had designed the game they were being asked about. The measures taken against this are described in Section 3.3. The comparative structure of the design also limits the effect on the central finding, because a general wish to be encouraging would raise accounts of both conditions rather than produce a consistent preference between them. It cannot be discounted for the absolute warmth of the accounts. Consistent with the reflexive orientation of the analysis [2], this relationship is named rather than treated as a neutral condition, and the accounts should be read with the awareness that they were given to a familiar interviewer.

5.3 Design Implications

The following implications are offered as propositions positioned within the existing conversation on designing emotionally adaptive games, rather than as isolated practitioner advice, indicating where the present findings extend, qualify, or complicate prior work.

Embed adaptation in atmosphere, not exposition. The consistent preference for atmospheric over text-based adaptation suggests that players experience adaptation more powerfully when it operates below the threshold of explicit awareness. The frameworks that catalogue adaptable content tend to

treat narrative and presentational channels as parallel and independent: Yannakakis and Togelius list "stories, dialogue, quests" alongside "music, camera, visual dynamics" without ranking their experiential force [34], and Nogueira et al. showed that atmospheric adaptation in a procedural horror game could modulate affect but did not test it against a textual alternative [24]. The present study suggests these channels are not experientially equivalent: adaptation delivered through the sensory layer was felt as more alive than the same narrative rendered through adaptive text. Read through Ermi and Mäyrä's distinction, sensory-level change did not merely add sensory immersion but deepened imaginative immersion [10], and the result gives empirical shape to Isbister's warning that emotionally designed content can "feel mechanical or manipulative at precisely the moment it succeeds technically" [17]. Making adaptation legible as text is one such moment. Designers may achieve greater emotional impact by investing adaptive effort in the atmospheric layer instead.

Reduce the system's voice. ELARA's presence in Condition A (Dialogue) made the system's adaptive behaviour visible, which invited analytical scrutiny rather than emotional engagement. This resonates with the Emote mode of the ACE framework, which positions emotional support as something a system enacts rather than announces [12], and it connects the finding to the placebo literature on perceived adaptivity. If, as Denisova and Cairns [9, 8] and Kosch et al. [20] demonstrate, the sense of being adapted to is substantially constructed by the user, then an explicit narrator that comments on its own adaptation risks puncturing that construction by drawing attention to the mechanism. Emotional adaptation may therefore be most effective when the system's presence is ambient rather than voiced.

Design for interpretive ambiguity. Players readily attributed intelligence and personal understanding to preauthored responses, which extends the placebo effect in games beyond its established domain. Where Denisova and Cairns located that effect in difficulty and Kosch et al. in the labelling of a system as AI-driven [9, 20], the present study locates it in narrative meaning: players read authored dialogue as personally directed understanding. The operative mechanism is the one Weizenbaum and Forer identified, responses general enough to be received as specific [33, 11]. Within the MDA framework, the implication is that the felt sense of being understood is produced at the level of dynamics and player interpretation rather than delivered by the mechanic [14]. Designers can therefore treat interpretive latitude as a resource to author for, rather than a deficit to engineer away through technically precise personalisation.

Consider repeated play as a design dimension. The within-subjects structure produced a player arc that was itself a finding: the evolution from exploration to strategic engagement or emotional deepening was not merely an order effect to be controlled for. This shift is legible through Calleja's account of incorporation, in which the game world is progressively internalised and the player's choices internalised by it [5], and through Bopp et al.'s finding that players actively construct meaning from emotionally charged play [1]. The affect-adaptive literature, focused as it is on single-session physiological response [7, 29], has largely treated repeated exposure as a confound to be controlled. The present findings suggest it can instead be a designed dimension: adaptive games that anticipate multiple playthroughs may structure each encounter as a distinct phase of a longer experiential trajectory.

5.4 Ethical and Societal Implications

The findings carry ethical weight beyond design practice. This study demonstrates that players readily attribute empathy, intelligence, and personal understanding to systems that possess none of these qualities, and that this attribution deepens rather than weakens emotional engagement. The same

mechanisms that make adaptive narratives meaningful, interpretive ambiguity, reduced system visibility, and atmospheric rather than explicit responsiveness, are also the mechanisms of emotional manipulation. The design implication to leverage the Barnum effect is therefore double edged: applied to a grief narrative in a consented research setting it produced meaningful experiences, but the same techniques deployed commercially, on emotionally vulnerable users, without transparency, raise genuine concerns. This is particularly acute for systems that simulate deceased or absent persons, a context Fabricade deliberately fictionalises. Participants in this study were protected by informed consent, content warnings, and a full debriefing that disclosed the preprogrammed nature of the adaptation. Commercial emotionally adaptive systems carry no such obligations. As adaptive and generative technologies spread, the boundary between responsive design and engineered emotional dependency will increasingly depend on designers' willingness to make the system's constructed nature legible, as Fabricade's own narrative ultimately does.

Framed as a question about games specifically, these concerns are not hypothetical, and the recent history of game regulation shows that they can become matters of law. Paid loot boxes, which use randomised rewards to encourage repeated spending, were judged to contravene gambling legislation in Belgium in 2018 and drew regulatory scrutiny across several European jurisdictions. The mechanism at issue there was financial: players were paying for chance-based rewards under conditions that resembled gambling. The mechanism examined in this study is emotional rather than monetary, but the underlying pattern is comparable. A designed system exploits a predictable psychological response, here the readiness to attribute understanding and intent to a responsive artefact, in order to intensify engagement. If games that shape spending behaviour through concealed psychological mechanisms have attracted regulation, it is reasonable to ask whether games that shape emotional attachment through equally concealed mechanisms should attract similar attention. The question resists an easy answer. Emotional engagement is the purpose of most narrative art, and a rule that treated all engineered feeling as suspect would condemn the medium itself. The distinction this study points to is transparency: Fabricade's effects were bounded because its constructed nature was ultimately disclosed, both through debriefing and through the narrative's own revelation that the memories were fabricated. A commercial system that simulates a responsive companion, particularly one modelled on a real or deceased person, and never discloses its mechanism, occupies very different ethical ground. Whether that ground should be governed by voluntary design norms, platform policy, or formal regulation of the kind now applied to loot boxes is a question this study raises but cannot settle.

5.5 Use of IT and AI Tools

Several software tools supported this thesis work. Fabricade was developed in Unity with the Ink narrative scripting language and deployed via itch.io. Sessions were conducted and recorded over Google Meet. Google Forms administered the consent form and questionnaire, and session logs were captured to Google Sheets. The thesis was written in LaTeX on Overleaf. AI based tools were used in two capacities, following the approach to generative AI in thesis work described by Johannesson [19]. Whisper was used for automated transcription of the recorded interview audio, with transcripts subsequently checked against the recordings. Claude (Anthropic) provided editorial support with the written text, including drafting suggestions, grammatical error correction, and tonal consistency. All AI generated output was critically reviewed, verified against the source data, and accepted or rejected

by the author. All coding, theme development, analysis, and interpretive decisions were carried out by the author.

5.6 Conclusion and Future Research

Several directions for future work emerge from the study's limitations and findings.

First, replication with different game genres and production contexts would test the transferability of the atmospheric-over-textual preference. The finding may be specific to text-based interactive fiction or may reflect a general principle about the experiential superiority of implicit over explicit adaptation.

Second, the boundary condition of emotional investment suggests that studies using participants' real personal data, as P7 proposed, could produce qualitatively different and potentially deeper engagement. This raises ethical considerations that would need to be carefully addressed.

Third, combining qualitative interviews with physiological measures (heart rate, galvanic skin response) would allow triangulation between what players report experiencing and what their bodies register. P7's session log data, 133.7 seconds of hesitation on a choice he later described as not emotionally affecting, suggests that self-report and physiological response may diverge in revealing ways.

Fourth, longitudinal designs involving multiple sessions over time could investigate whether the player evolution documented in Theme 4 continues to develop or reaches a plateau. The present study captured a single arc across two playthroughs. Whether adaptive games sustain engagement over longer trajectories remains an open question.

This study set out to understand how players experience emotionally adaptive game narratives when the mode of adaptation varies. It does not offer a complete answer, but it supports a clear partial one. Players did not experience adaptation as a technical property of the system but as an interpretive encounter, and the mode of adaptation shaped the character of that encounter. Adaptation embedded in atmosphere, operating below the threshold of explicit awareness, was experienced as more alive and more personal than adaptation delivered through explicit narrative commentary, even though the latter carried more semantically adaptive content. Across both modes, players constructed the sense of being understood rather than receiving it from the system, attributing intelligence and intent to preauthored responses. And in both modes, emotional depth was conditional on the player's investment in the fictional relationship rather than on the sophistication of the adaptation. Taken together, these findings suggest that the emotional force of adaptive narrative lies less in the accuracy of the system's response than in the interpretive work the player brings to it. The adaptation is authored. The empathy is made by the player, not delivered by the system. This is a partial answer, grounded in a single prototype and a small sample, but it reframes emotionally adaptive design as a matter of interpretation rather than technical precision. If players supply the empathy that adaptive systems appear to offer, then the central task for design, and the central question for the research that follows this study, is not how to make systems understand players better, but how to make room for the understanding that players prove so willing to create.

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Appendix A: Consent Form

The following information and consent form was provided to all participants before the session. Consent was collected digitally via Google Forms; remote participants provided their name and date in lieu of a physical signature.

Informed Consent Form

Study Title: Algorithmic Empathy: Exploring Emotionally Adaptive Game Narratives
Researcher: Sakib Ahsan Dipto, Master's Programme in Design for Creative and Immersive Technology, Department of Computer and Systems Sciences (DSV), Stockholm University
Supervisor: Donald McMillan, DSV, Stockholm University

Purpose of the Study

This study investigates how different forms of adaptive design in a narrative game affect the player's emotional experience. You will play a short text based narrative game that explores themes of loss and grief, and then participate in an interview about your experience.

What Participation Involves

- Playing a narrative game twice (two different versions), approximately 15 to 20 minutes each
- A semi structured interview afterward, approximately 15 to 20 minutes
- The session will be screen recorded and audio recorded for research purposes

Important Information About Content

The game explores themes of grief, loss, and bereavement. The content is fictional but may evoke emotional responses. Please consider your own wellbeing before agreeing to participate. You may stop at any time without explanation.

Your Rights

- Participation is entirely voluntary
- You may withdraw at any time during or after the session without giving a reason
- If you withdraw, your data will be deleted
- You may decline to answer any interview question

Data Handling

- Audio and video recordings will be used solely for transcription and analysis
- Your identity will be anonymised in the thesis; no real names or identifying details will be published
- All data will be stored securely and deleted after the thesis is completed and graded
- The behavioural data logged by the game (choices, response times) is stored locally and anonymised

Consent

By signing below, you confirm that:

- You have read and understood the above information
- You have had the opportunity to ask questions
- You consent to participate voluntarily in this study
- You consent to the session being recorded for research purposes

Name: _____

Signature: _____

Date: _____

For remote participants: You may provide your name and date via email or digital form in lieu of a physical signature.

Appendix B: Interview Questions

The semi structured interview followed the guide below. Questions were asked flexibly: probes were used only where relevant, questions already answered naturally during conversation were skipped, and participants were allowed to speak without interruption. Pre recorded video clips of key adaptive moments (for Condition A, ELARA's dialogue tone shifting after a defiant versus an accepting choice; for Condition B, the text glitch effect, the screen glitch at the hospital corridor threshold, and the vignette and glow shift) were screen shared during the video prompted recall section.

Opening

1. How would you describe your overall experience of the game?
    - Probe: "Can you say more about that?"
    - Probe: "What stood out to you most?"

On Emotional Engagement

2. Did the game affect you emotionally in any way? Can you describe that?
3. Was there a moment that stayed with you more than others? What was it about that moment?

On Adaptation and Perception

4. Did you notice anything changing or shifting during either session?
    - Probe: "What gave you that impression?"
5. Did the game ever feel like it was responding to you personally?
    - Probe: "Can you describe what that felt like?"

Video Prompted Recall

After each pre recorded clip was shown:

- What was going through your mind when this happened?
- How did this moment feel to you?

If a participant did not remember the moment, this was noted as data; the researcher did not explain what was happening in the clip.

On Agency and Choice

6. When you were making choices, what were you thinking about?
7. Did your choices feel like they had weight, like they were doing something in the game?
    - Probe: “Was there a choice you hesitated on? What was going through your mind?”
8. Was there a moment where you wished you had chosen differently?

On ELARA

9. What did you make of ELARA as a presence in the game?
    - Probe: “Did ELARA feel different between the two sessions?”

On the Comparison Between Conditions

10. Were the two sessions different to you as an experience? How would you describe that difference?
11. In which session did your choices feel more meaningful? What made it feel that way?
12. Which session felt more alive to you? How would you describe that difference?

Closing

- Is there anything about your experience that you want to add, anything we have not talked about that felt important to you?

Appendix C: Session Log Structure and Summary

Fabricade wrote a structured log for every session. Each log is a time ordered list of events, and each event records a timestamp, the elapsed session time in seconds, an event type (`scene_enter`, `choice_presented`, `choice_made`, or `story_end`), the scene in which it occurred, a detail string, and the hesitation time in seconds between a choice being presented and selected. The final `story_end` event records the closing value of every internal Ink variable together with a labelled record of the six choices made.

Table 1 summarises the closing state of all 20 sessions. Participants are pseudonymised, and hesitation figures exclude the untimed `Continue` prompts. Emotional posture is unbounded in the implementation and is reported as recorded.

P	Cond	Dur (s)	Med. hes	Max hes	Open	Defl	Resist	Posture	Trust
P1	A	397	34	78	2	0	1	2	0
P1	B	220	18	33	0	1	2	-4	1
P2	A	577	44	100	1	1	1	1	1
P2	B	515	49	85	0	2	0	-1	0
P3	A	450	35	80	1	1	1	0	0
P3	B	251	20	34	1	2	1	0	-1
P4	A	448	32	69	0	3	1	-2	0
P4	B	438	30	54	2	1	1	2	0
P5	A	489	46	65	0	2	0	0	0
P5	B	301	22	73	0	1	2	-4	2
P6	A	334	25	64	0	2	0	0	1
P6	B	344	29	40	0	1	2	-4	2
P7	A	758	60	134	1	1	2	-2	0
P7	B	473	39	80	2	0	2	1	-1
P8	A	410	30	87	1	2	0	2	-2
P8	B	254	25	34	2	2	0	4	-2
P9	A	418	26	90	1	1	1	-1	1
P9	B	315	20	61	1	2	1	1	-2
P10	A	403	39	63	1	2	2	-1	2
P10	B	278	29	44	1	4	0	2	-1

Table 1: Closing state of all 20 Fabricade sessions, by participant and condition. Dur is total session duration, Med. hes and Max hes are the median and maximum hesitation times across the seven narrative decision points, and the remaining columns are the closing values of the internal Ink variables.

The raw event streams were retained in full and were used to locate the specific moments participants described in their interviews, in particular the pause at the revelation scene reported in Chapter 4.